SECONDARY GEAR AND OUTPUT GEAR END PLAY MEASUREMENT/ADJUSTMENT

id051700670000

Preparation Before Servicing

1. Print out the measurement/adjustment value input sheet. (See MEASUREMENT/ADJUSTMENT VALUE INPUT SHEET.)

Note

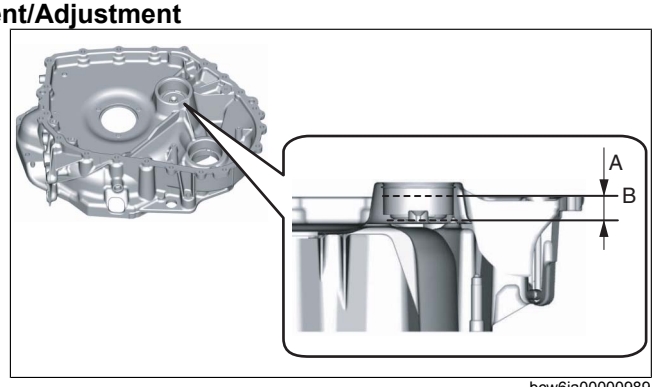
- When performing the measurement/adjustment, input the measured and calculated values into the measurement/adjustment value input sheet.
- If the measurement/adjustment value input sheet has already been printed out for the other measurements/adjustments, use the sheet.

Secondary Gear and Output Gear End Play Measurement/Adjustment

1. Calculate distance A of the converter housing using the following procedure:

A : Distance A
B : Converter housing end (alignment surface with transaxle case)

(1) Remove any remaining old sealant on the contact surfaces of the transaxle case and converter housing.



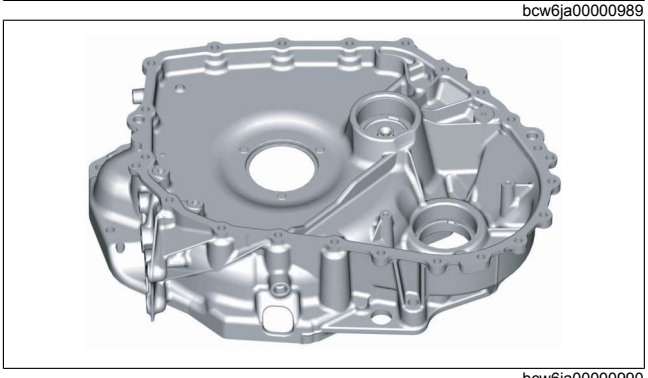
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(2) Set the transaxle case of the converter housing so that the mating surface side is facing upward.
(3) Measure the thickness of the straight edge rulers.

Note

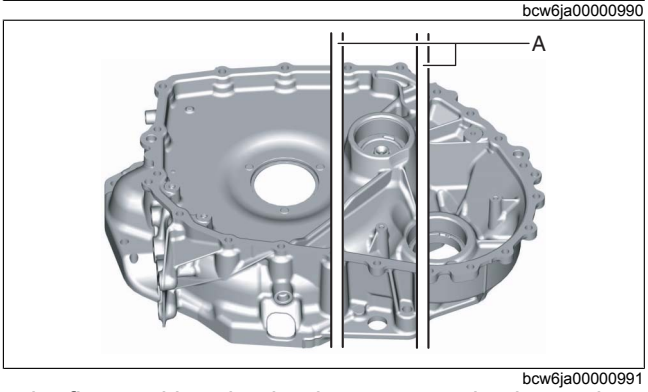
- Recommended measuring instrument: Micrometer

(4) Input the straight edge ruler thickness into the measurement/adjustment value input sheet.


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(5) Set 2 straight edge rulers and the depth micrometer to the converter housing.

A : Straight edge ruler



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(6) Measure the optional positions (3 locations) shown in the figure with a depth micrometer and subtract the thickness of the straight edge ruler to calculate distance A.

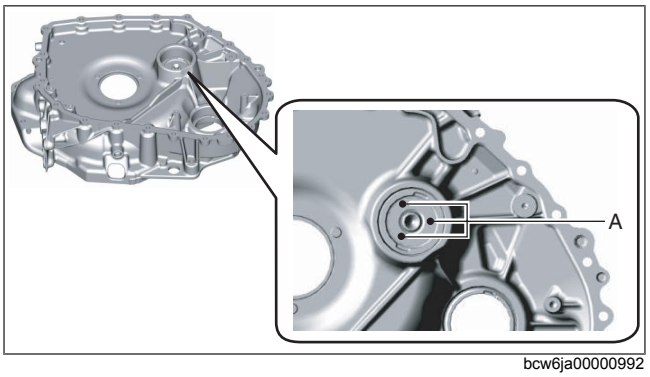
Note

- Recommended measuring instrument: Depth micrometer, straight edge ruler

A : Measurement location

(7) Input distance A and distance A average value into the measurement/adjustment value input sheet.

(8) Remove the 2 straight edge rulers and depth micrometer from the converter housing.

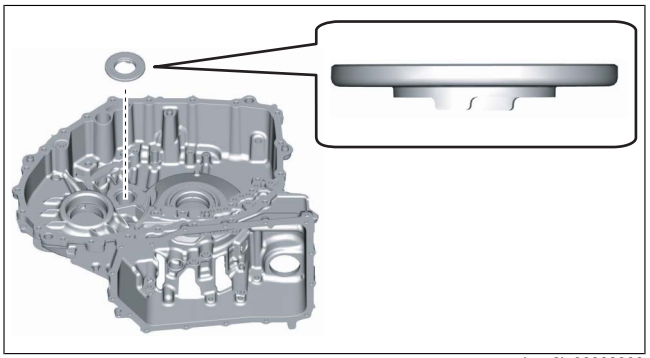


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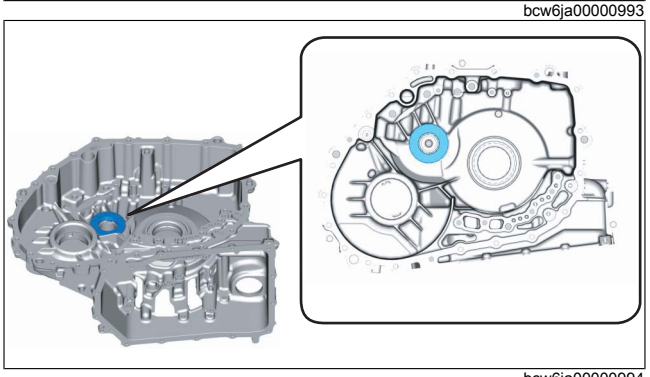
2. Assemble the thrust needle bearing.

Note

- Thrust needle bearing size: Outer diameter approx. 67 mm {2.6 in}

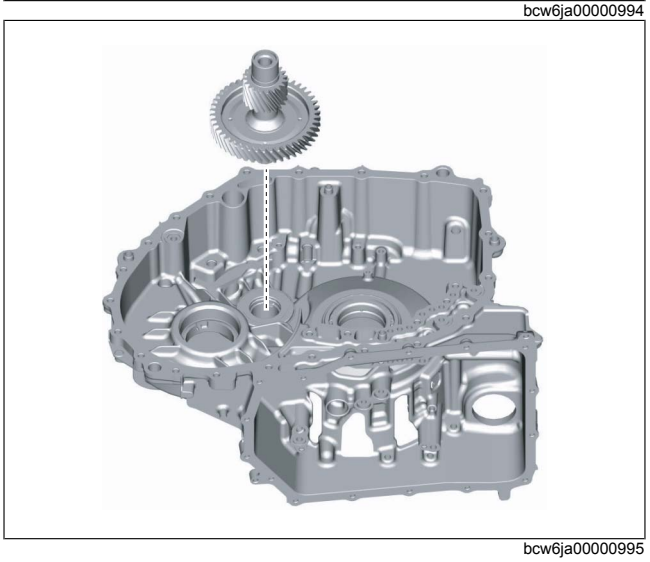


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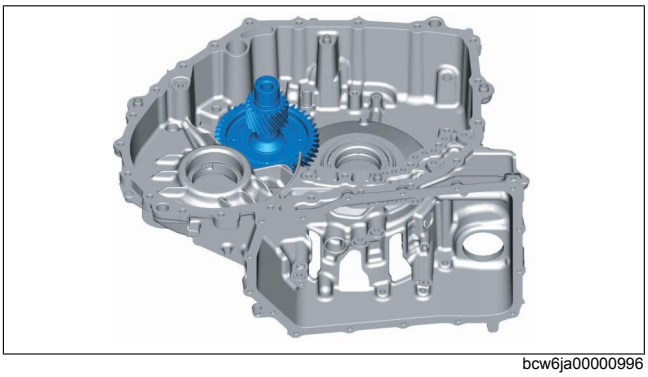


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3. Assemble the secondary gear and output gear.

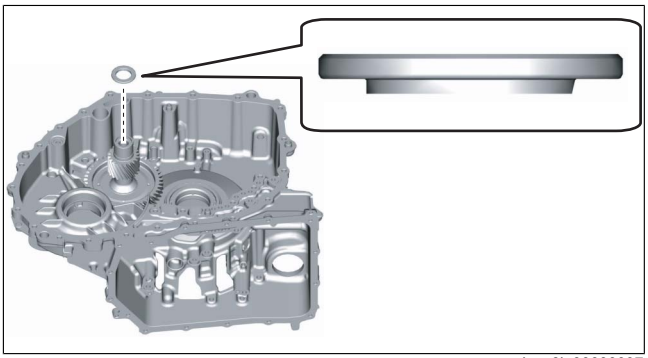


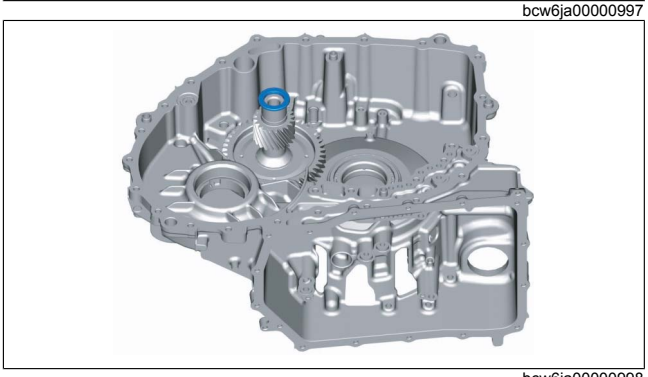
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4. Assemble the thrust needle bearing.

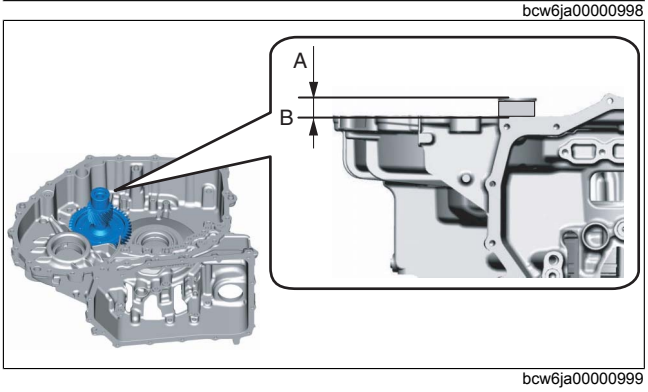
Note
- Thrust needle bearing size: Outer diameter approx. 39.4 mm {1.55 in}


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5. Calculate distance B of the secondary gear and output gear using the following procedure.

A : Distance B
B : Transaxle case end (alignment surface with converter housing)



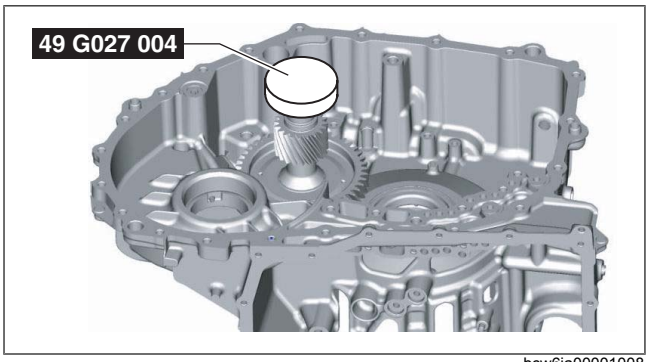
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(1) Measure the thickness of the SST (49 G027 004).

Note
- Recommended measuring instrument: Micrometer

(2) Input the thickness of the SST in the measurement/adjustment value input sheet.
(3) Install the thickness measured SST on to the thrust needle bearing.

Caution
- **Install the SST to the center of the thrust needle bearing to reduce measurement differences on the secondary gear and output gear end play measurement.**

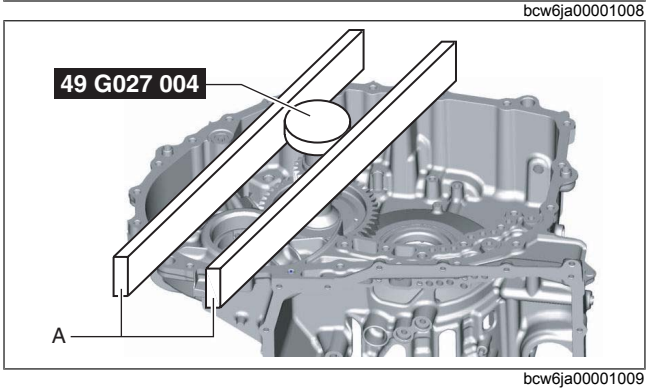


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(4) Set the 2 straight edge rulers and depth micrometer to the transaxle case.

A : Straight edge ruler

(5) Measure near the SST center with a depth micrometer.



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Caution

- **Measure near the center of the SST to reduce measurement differences on the secondary gear and output gear end play measurement.**

Note

- Recommended measuring instrument: Depth micrometer, straight edge ruler

(6) Input the depth micrometer measurement value into the measurement/adjustment value input sheet.

(7) Perform the following calculation to calculate distance B.

Distance B = A - D - E

A: Straight edge ruler thickness
D: SST thickness
E: Depth micrometer measurement value

Note

Example

A: Straight edge ruler thickness is 39.76 mm {1.565 in}
D: SST thickness is 15.00 mm {0.5906 in}
E: Depth Micrometer measurement value is 5.82 mm {0.229 in}

Distance B = 39.76 mm {1.565 in} - 15.00 mm {0.5906 in} - 5.82 mm {0.229 in} = 18.94 mm {0.7457 in}

(8) Input distance B into the measurement/adjustment value input sheet.

6. Select the appropriate shim from the following table.

Range*	Shim thickness
Exceeds 1.644 mm {0.06472 in}, 1.694 mm {0.06669 in} or less	1.45 mm {0.0571 in}
Exceeds 1.594 mm {0.06276 in}, 1.644 mm {0.06472 in} or less	1.40 mm {0.0551 in}
Exceeds 1.544 mm {0.06079 in}, 1.594 mm {0.06276 in} or less	1.35 mm {0.0531 in}
Exceeds 1.494 mm {0.05882 in}, 1.544 mm {0.06079 in} or less	1.30 mm {0.0512 in}
Exceeds 1.444 mm {0.05685 in}, 1.494 mm {0.05882 in} or less	1.25 mm {0.0492 in}
Exceeds 1.394 mm {0.05488 in}, 1.444 mm {0.05685 in} or less	1.20 mm {0.0472 in}
Exceeds 1.344 mm {0.05291 in}, 1.394 mm {0.05488 in} or less	1.15 mm {0.0453 in}
Exceeds 1.294 mm {0.05094 in}, 1.344 mm {0.05291 in} or less	1.10 mm {0.0433 in}
Exceeds 1.244 mm {0.04898 in}, 1.294 mm {0.05094 in} or less	1.05 mm {0.0413 in}

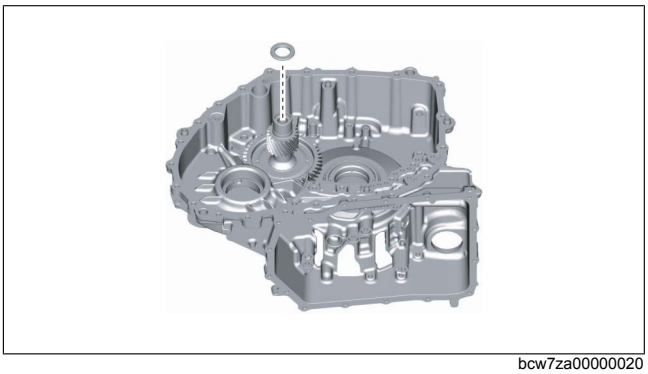
Range*	Shim thickness
Exceeds 1.194 mm {0.04701 in}, 1.244 mm {0.04898 in} or less	1.00 mm {0.0394 in}
Exceeds 1.144 mm {0.04504 in}, 1.194 mm {0.04701 in} or less	0.95 mm {0.037 in}
Exceeds 1.094 mm {0.04307 in}, 1.144 mm {0.04504 in} or less	0.90 mm {0.035 in}
Exceeds 1.044 mm {0.04110 in}, 1.094 mm {0.04307 in} or less	0.85 mm {0.033 in}
Exceeds 0.994 mm {0.0391 in}, 1.044 mm {0.04110 in} or less	0.80 mm {0.031 in}
Exceeds 0.944 mm {0.0372 in}, 0.994 mm {0.0391 in} or less	0.75 mm {0.030 in}
Exceeds 0.894 mm {0.0352 in}, 0.944 mm {0.0372 in} or less	0.70 mm {0.028 in}
Exceeds 0.844 mm {0.0332 in}, 0.894 mm {0.0352 in} or less	0.65 mm {0.026 in}
Exceeds 0.794 mm {0.0313 in}, 0.844 mm {0.0332 in} or less	0.60 mm {0.024 in}
Exceeds 0.744 mm {0.0293 in}, 0.794 mm {0.0313 in} or less	0.55 mm {0.022 in}
Exceeds 0.694 mm {0.0273 in}, 0.744 mm {0.0293 in} or less	0.50 mm {0.020 in}

* : The range is the sum of the average value of distance A subtracted from distance B.

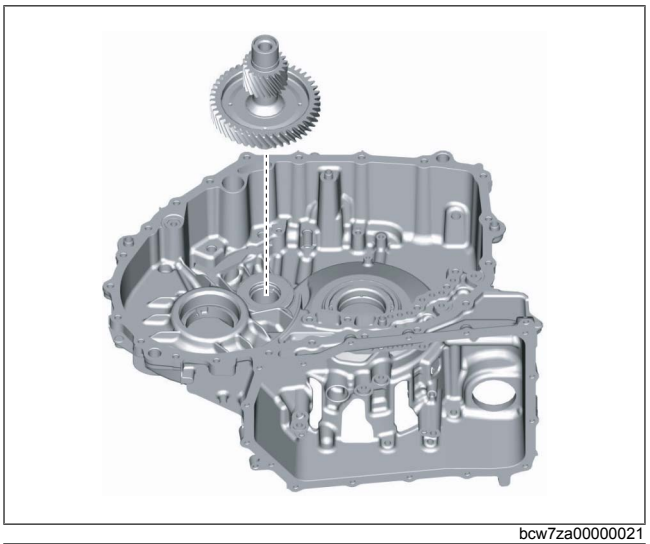
Range = C - F
C: Average value of distance A
F: Distance B

Note
Example
C: Average value of distance A is 20.15 mm {0.7933 in}
F: Distance B is 18.94 mm {0.7457 in}
Range = 20.15 mm {0.7933 in} - 18.94 mm {0.7457 in} = 1.21 mm {0.0476 in}, a shim of 1.00 mm {0.0394 in} thickness should be selected.

7. Remove the 2 straight edge rulers and depth micrometer from the transaxle case.
8. Remove the SSTs.
9. Remove the thrust needle bearing.

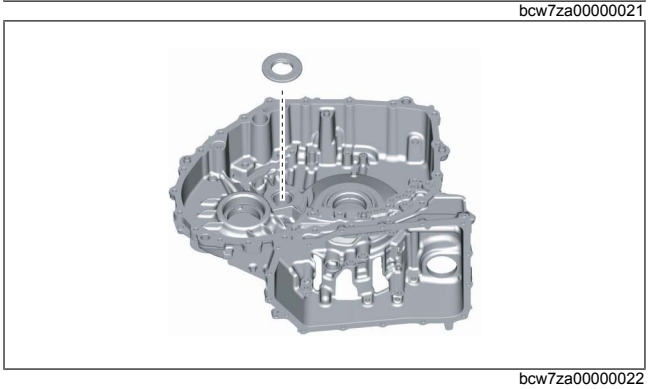

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10. Remove the secondary gear and output gear.



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11. Remove the thrust needle bearing.



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